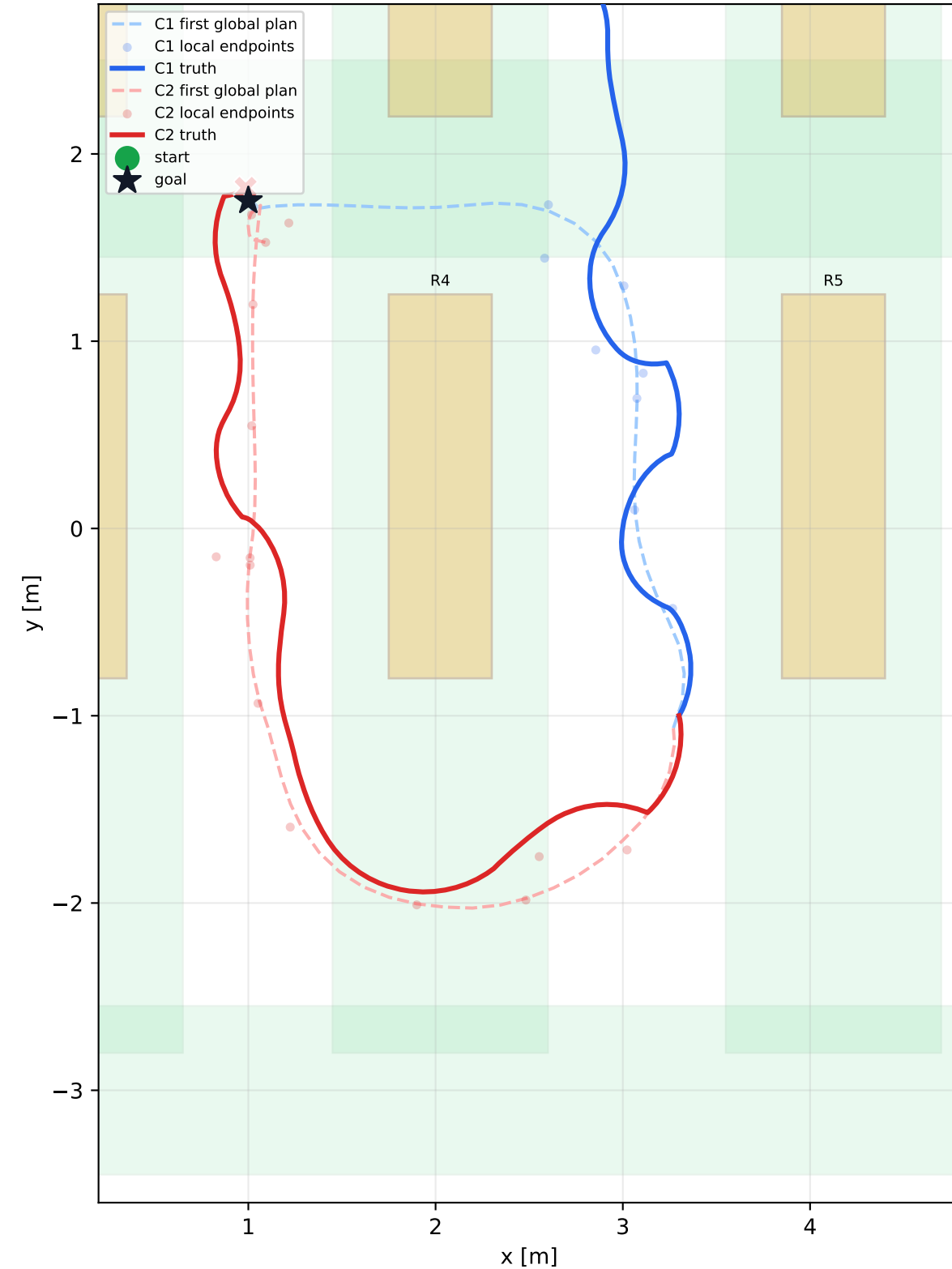


F45 - Tracking yaw + local solver timing diagnostic

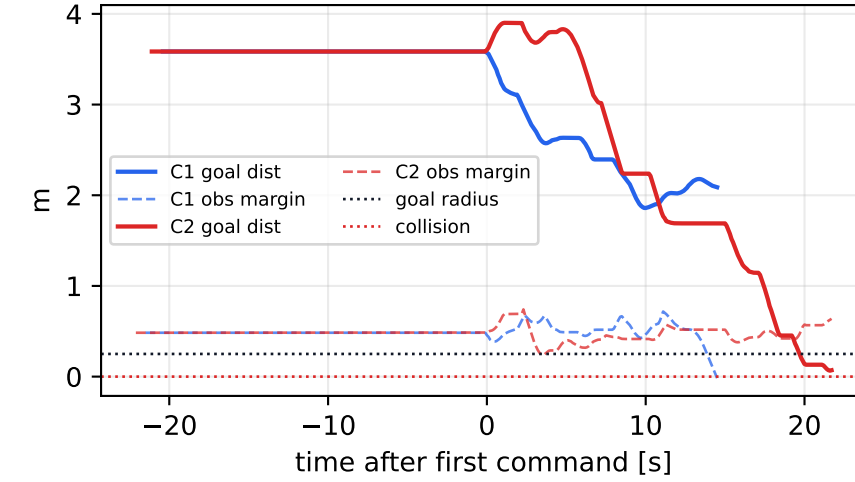
R4

R5

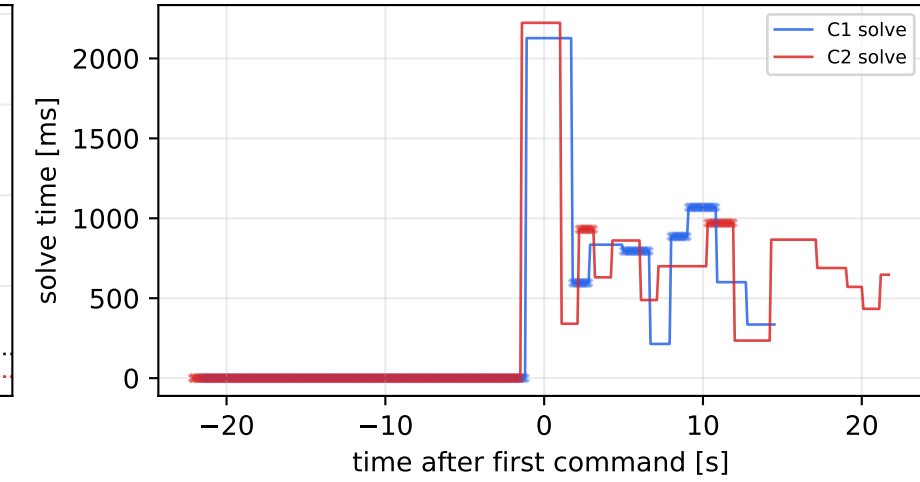
(a) First route, local endpoints, executed trajectory



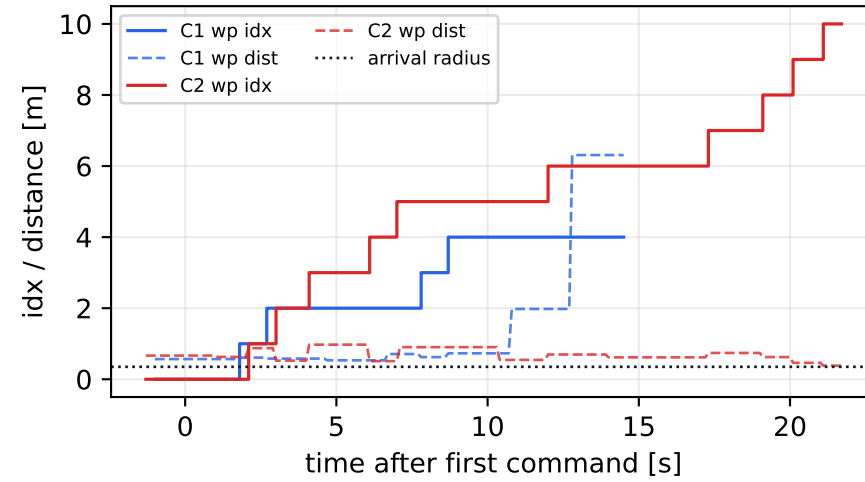
(b) Goal progress and obstacle margin



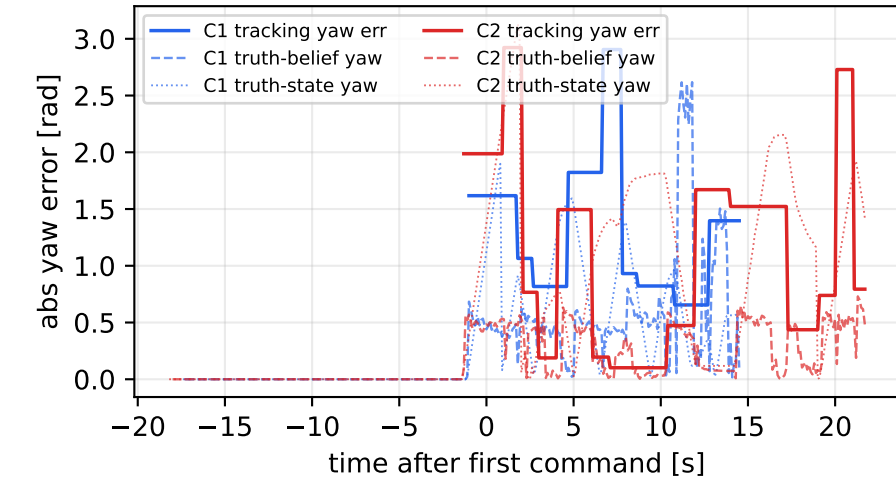
(c) Local solve time (x = non-success)



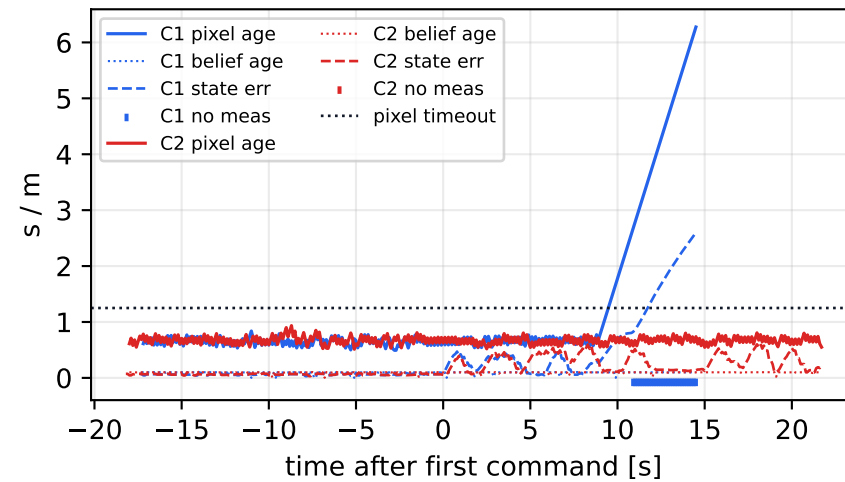
(d) Waypoint progress



(e) Heading/yaw errors



(f) Perception freshness and localization



F45 controlled tracking-yaw smoke

C1: collision, path=5.10m, min_goal=1.86m
 err=0.342m, min obs=-0.017m
 solve μ /p90=829/1280 ms, 23 ms/eval
 maxiter/non-success rows=16%
 yaw track μ /p90=1.30/1.82 rad
 pixel age p90/max=3.11/6.28s

C2: goal_reached, path=6.29m, min_goal=0.07m
 err=0.187m, min obs=0.239m
 solve μ /p90=756/959 ms, 24 ms/eval
 maxiter/non-success rows=6%
 yaw track μ /p90=1.13/1.99 rad
 pixel age p90/max=0.76/0.94s

Interpretation:

1. Spawn yaw is clean; yaw problems are runtime/tracking, not init.
2. Global plans still choose the expected route families.
3. Local H15/maxiter20 is too budget-starved: frequent maxiter hits.
4. C1 additionally suffers stale camera belief before collision.
5. C2 localizes better but still collides: local tracking/clearance remains failing